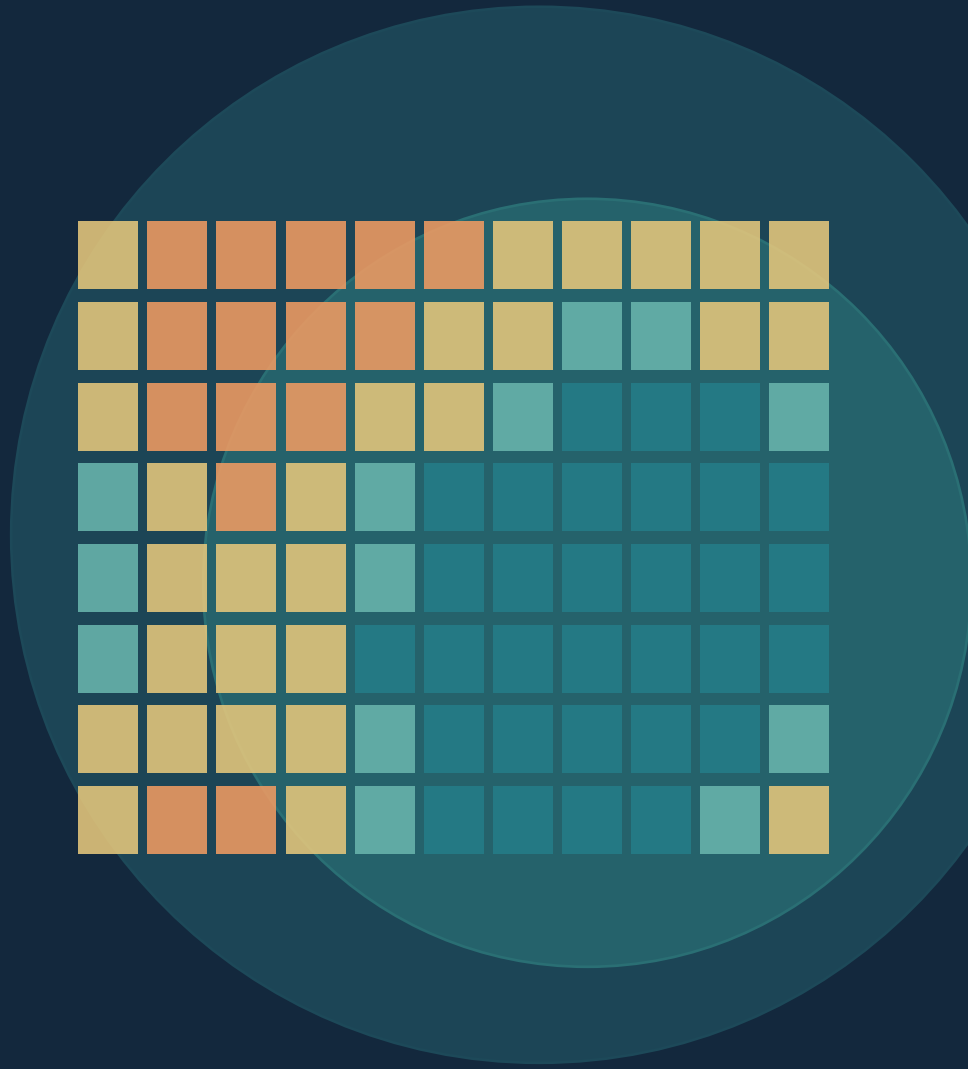
FIRST CHECKPOINT / PROJECT PROPOSAL

Foundation Models for High-Resolution Air Quality Maps

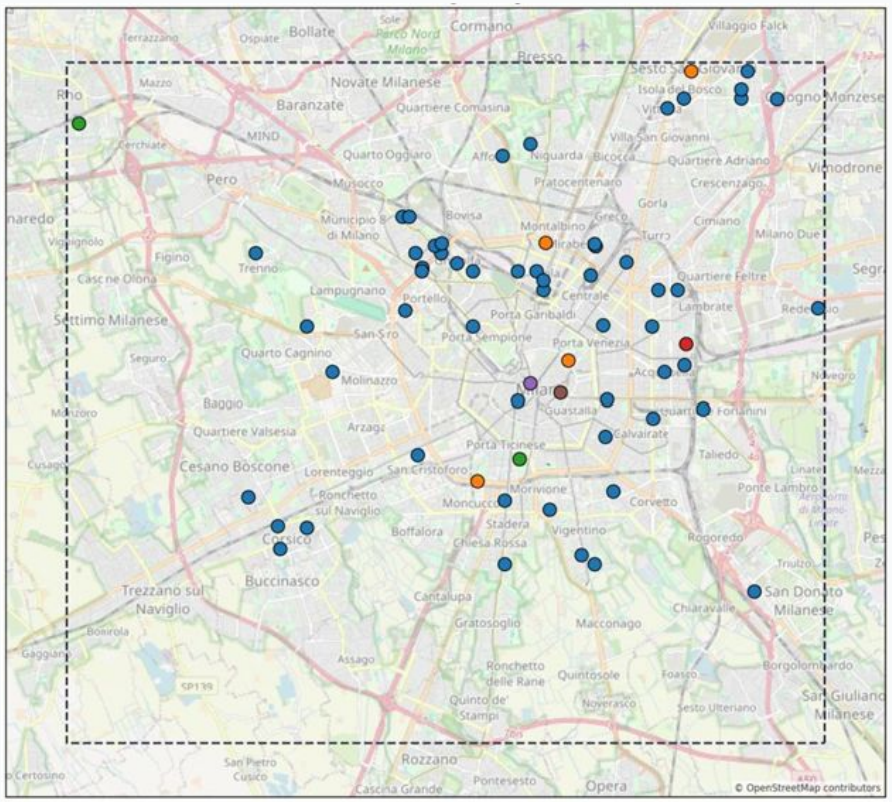


Urban-scale pollution nowcasting

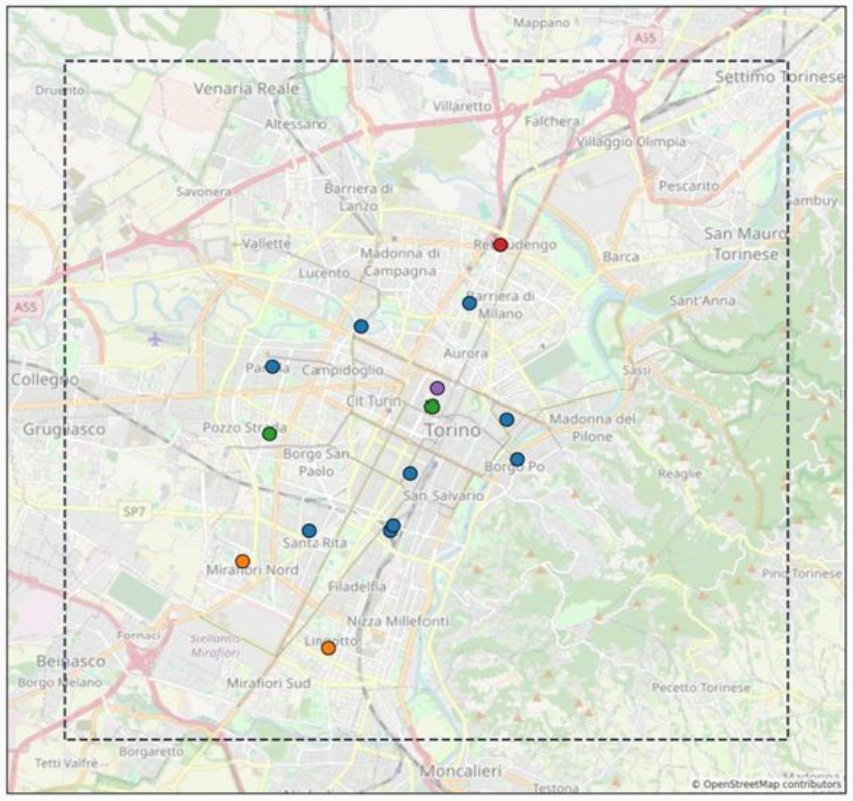
Applied Data Science Project • Turin / Milan

Seyedmostafa Seyedi - s352883
Cristian Sician - s362986

The problem: stations measure points, not entire cities



MILAN • 65 STATIONS



TURIN • 15 STATIONS

VALUE PROPOSITION

From sparse measurements to healthier, smarter cities

Daily ~1 km pollution maps can reveal local patterns that station-only views miss.

01

Detect hotspots

Identify areas of elevated pollution and urban–rural gradients.

02

Informed decisions

Support mobility measures, monitoring networks and exposure assessment.

03

Improve monitoring

Help environmental agencies understand which areas are not sufficiently monitored

VALUE PROPOSITION

High-resolution air-quality maps turn sparse observations into actionable information for healthier, more sustainable cities.





Innovation & Infrastructure

Foundation models and Earth observation data enable innovative, scalable urban air-quality mapping.



Sustainable Cities

Neighborhood-scale pollution maps help cities identify hotspots and guide better urban and mobility planning.



Climate Action

Improved environmental monitoring supports evidence-based decisions to address urban pollution challenges.

Project objectives and scope

TASK	Urban-scale concentration regression for 5 pollutants
POLLUTANTS	NO_2 • SO_2 • O_3 • PM10 • PM2.5
MAP	Predict pollutant concentration maps for the input day
SPATIAL RESOLUTION	Target ≈1 km
GEOGRAPHY	Turin and/or Milan; daily temporal resolution

STAKEHOLDERS

Who contributes data — and who can use the results

Environmental agencies

ARPA and regional monitoring bodies · primary operational users

Municipalities

City planning, mobility and low-emission-zone teams

Public-health authorities

Exposure assessment and risk communication

Academic researchers

Model benchmarking and urban digital-twin studies

Data & project partners

Copernicus / ESA data ecosystems, LINKS mentors

USER PERSONAS

People who benefit from neighborhood-scale air-quality maps

Environmental analyst - Sara

CHALLENGE Needs validated daily estimates between sparse monitoring stations.

BENEFIT Identify hotspots and guide sensor placement.

Municipal planner - Nora

CHALLENGE Needs local evidence to assess mobility and emissions initiatives.

BENEFIT Compare patterns across neighborhoods and days.

Public-health officer - Giovanni

CHALLENGE Needs spatial context for population exposure estimates.

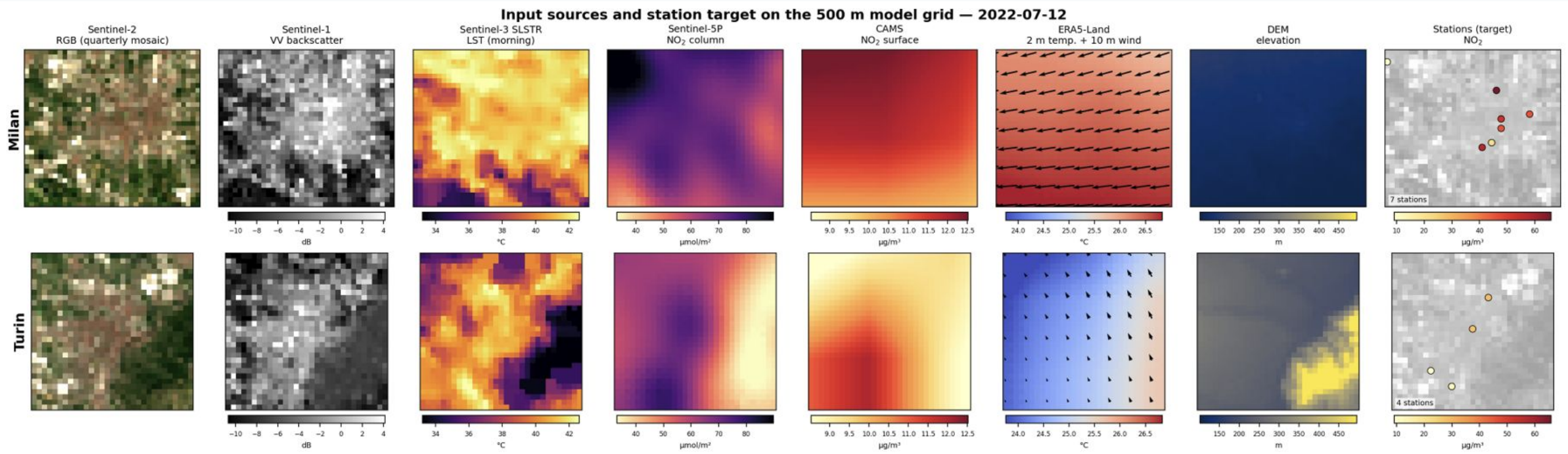
BENEFIT Support communication and health alerts.

Air-quality researcher - Josh

CHALLENGE Needs reproducible inputs, predictions and metrics.

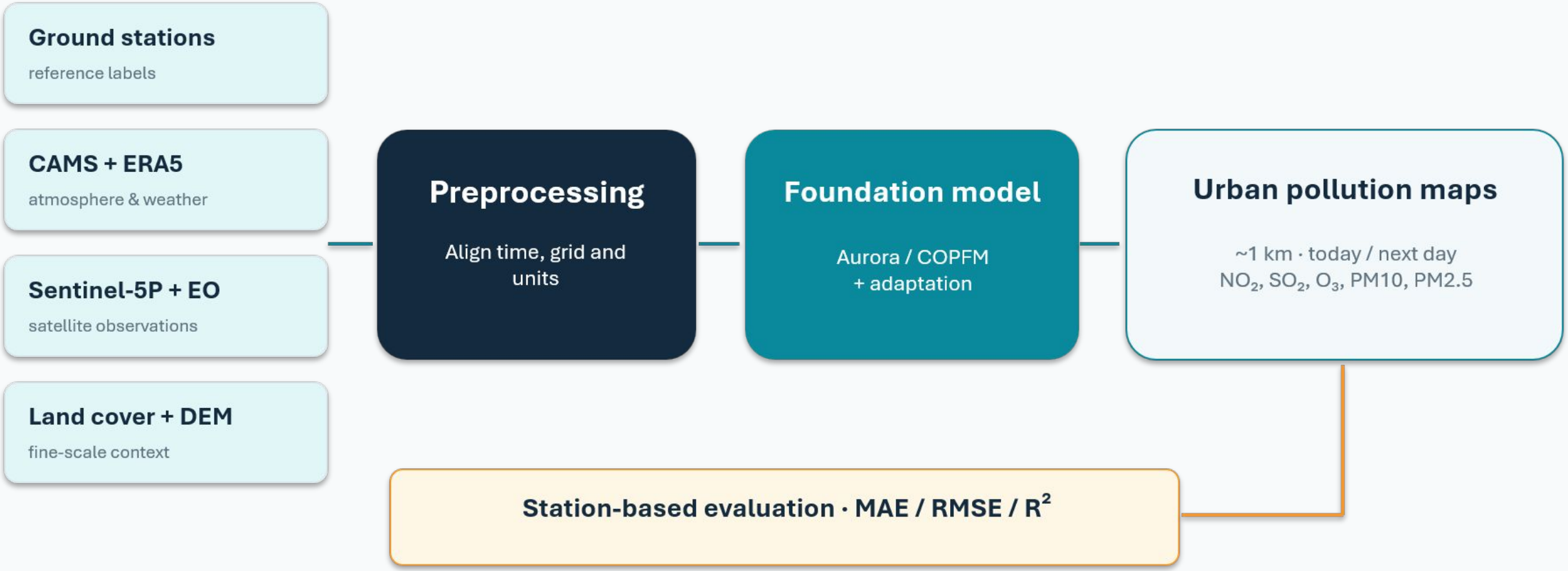
BENEFIT Compare models with independent validation.

Available data: complementary views of the same city



FUNCTIONAL DIAGRAM

Multi-source data → foundation model → high-resolution maps



Evaluation: both numerical accuracy and map quality

MAE

Lower is better

Average absolute prediction error

RMSE

Lower is better

Penalizes larger errors more strongly

R^2

Higher is better

Explained variability on unseen observations

Validation protocol

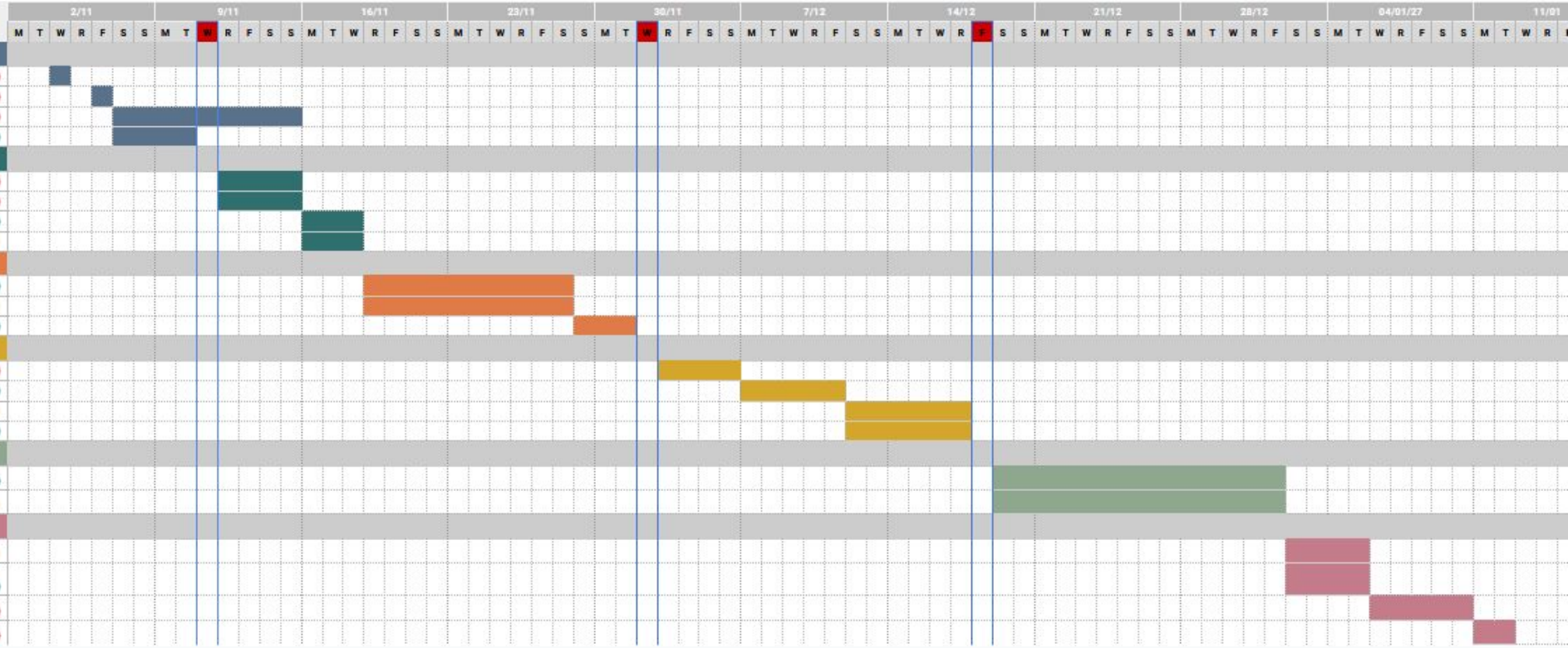
- Hold out time periods
- Compare against the coarse baseline at actual station locations
- Inspect realistic road/industrial hotspots and urban–rural gradients

GANTT Chart

WPs	Lead Name	November	December	January
Project Management	C. Sician			
Research and Familiarization	M. Seyedi			
Data Exploration and Visualization	C. Sician			
Model implementation and Training	M. Seyedi			
Evaluation and Testing	C. Sician			
Report and Documentation	M. Seyedi			

GANTT / PLANNING

WBS NUMBER	TASK TITLE	PROGRESS	TASK OWNER
1	**Project Management**		
1.1	Meeting with the mentor		Team
1.2	Define Project Objectives		Team
1.3	Project Planning		Team
1.4	Preparing the Project Presentation		Mostafa
2	**Research and Familliarization**		
2.1	Paper Reading		Team
2.2	Lab Setup		Team
2.3	Aurora		Mostafa
2.4	Copernicus		Cristian
3	**Data Exploration and Visualization**		
3.1	Exploring the Available Datasets		Mostafa
3.2	Visualize Data		Cristian
3.3	Split the Data into Training/Testing		Mostafa
4	**Model Implementation and Training**		
4.1	Understand model architecture		Team
4.2	Implement and verify the baseline model		Mostafa
4.3	Train or fine-tune the model head		Cristian
4.4	Validate the trained model		Mostafa
5	**Evaluation and Testing**		
5.1	Find the Best Combination of Parameters		Mostafa
5.2	Testing the Model based on Different Metrics		Cristian
6	**Report and Documentation**		
6.1	Gathering Results		Cristian
6.2	Document the Final Results and Preparing the Report		Mostafa
6.3	Prepar Final Presentation Slides		Team
6.4	Reveiw and Submit Documentaion		Team



OUTPUTS

Expected Deliverables

01

Reproducible benchmark

Compare Aurora and COPFM with appropriate baselines and station-based metrics.

02

High-resolution maps

Produce daily ≈1 km concentration maps over Turin and/or Milan.

03

Public GitHub repository

THANK YOU

Questions & Discussion

Foundation Models for High-Resolution Air Quality Maps

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